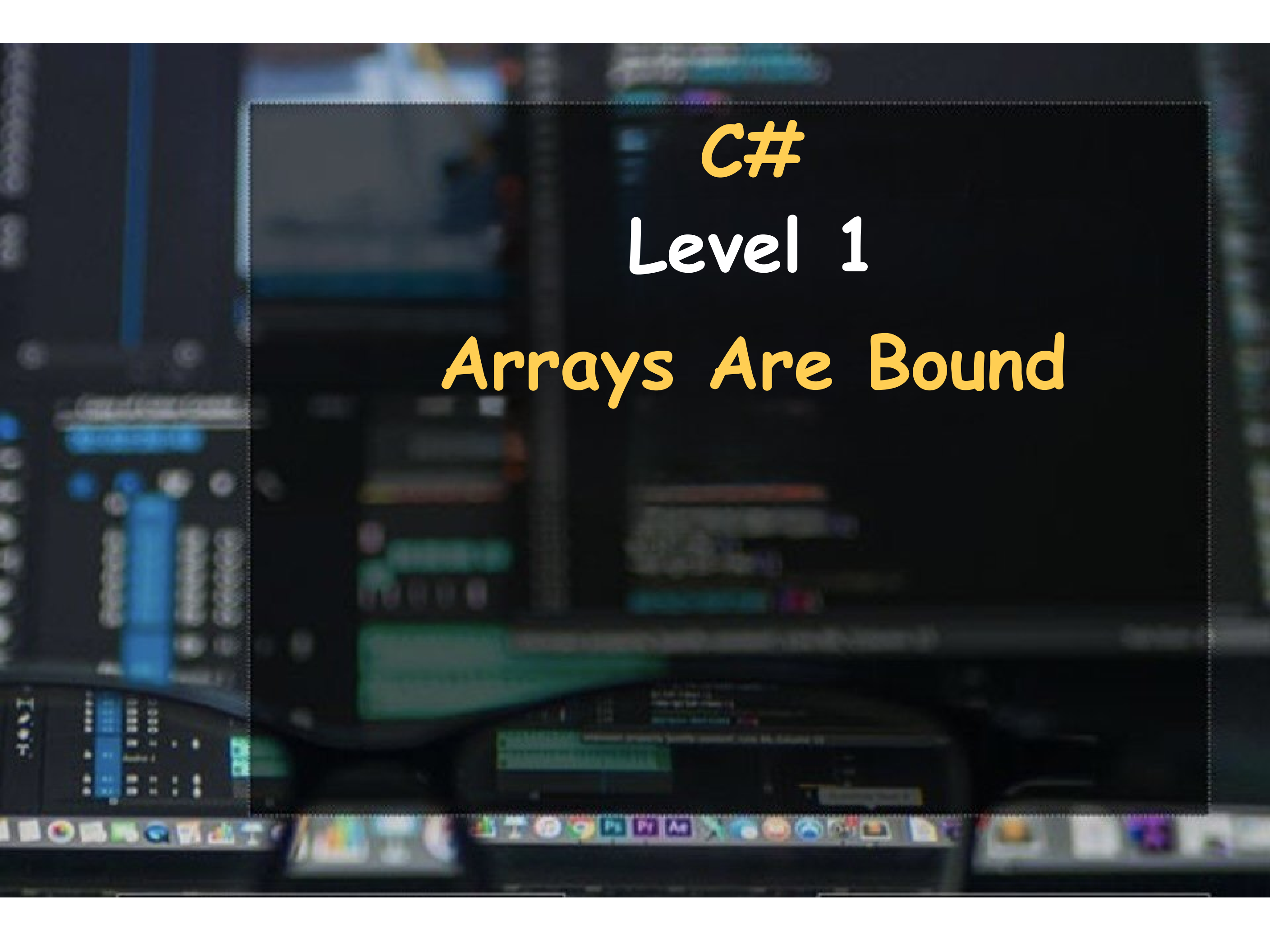


The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). On the left side, there's a sidebar with some icons and text. At the bottom, a Windows taskbar is visible with several application icons like Chrome, Photoshop, and others.

C#

Level 1

Arrays Are Bound



C#

Level 1

Arrays Are Bound

Definition

Fixed Array Size

-  A *C#* array has a fixed length after it is created.
-  Elements can only be accessed within the valid index range.

Explanation

Array Indexes

-  Arrays are zero-indexed, so the first element has index 0.

-  The last valid index is $\text{Length} - 1$.

💡 Example

`int[] numbers = { 10, 20, 30, 40, 50 };`

`Length = 5`

`Valid indexes = 0 → 4`

Note

- ◆ An array with 5 elements has indexes 0 through 4.
- ◆ Index 5 is outside the valid range.



Comparison



C#

- ✓ Normal array indexing is bounds-checked at runtime.
- ✓ Invalid indexes throw `IndexOutOfRangeException`.



C++

- ✓ `operator[]` does not perform bounds checking.
- ✓ Out-of-bounds access results in undefined behavior.

✦ Summary

- ✓ Arrays have a fixed length after creation.
- ✓ Indexes start from 0.
- ✓ The last valid index is Length - 1.
- ✓ Invalid indexing in C# throws `IndexOutOfRangeException`.

Thank you 😊